

Reading the Data: An Analysis of Willamette University Library's E-Resource Usage

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Abstract

Willamette University's Library has a host of online journals from a handful of publishers. As Willamette changes and the library reevaluates its needs, the librarians need to consider which electronic materials to maintain and which to let go. With no recent way to analyze the usage patterns of journals and articles, we performed analysis on usage data in tandem with enrollment data to make actionable suggestions on how to fairly maintain the available online resources. From our analysis, we have found that journal usage ebbs when school is not in session, and that the Spring and Fall terms generally have higher usage, with Fall having the most consistent patterns of high usage. In the most recent school year, usage has decreased by 29%.

Introduction and Motivation

Willamette University has 3 libraries: Mark O. Hatfield Library (Salem Main Campus), J.W. Long Law Library (Salem Law Campus), and the Albert Solheim Library (PNCA Portland Campus). Each has its own physical resources and shares a collection of electronic resources across all libraries. This includes journals, research papers and articles, and often comes in packages bought by the library from publishers. In recent years, the Willamette Librarians have had no avenue to analyze patterns in journal requests and usage.

The library staff desires to understand the Willamette community's online journal usage as a whole as well as how it relates to enrollment in different departments. Through our analysis, the librarians will be able to consider which electronic materials are the most important to maintain, and which to deprioritize. Since some departments are smaller, they may be more likely to be hit by cuts if their resources have lower usage request figures. Willamette's librarians are striving to optimize their resources while maintaining fairness between departments and being good stewards of research.

Research Questions

RQ1: What are the patterns in usage with regard to subject matter, departments, semesters, and classes?

RQ2: Can we use the patterns in usage to make suggestions on which resources to prioritize, and which resources are not as important?

Impact & stakeholders:

Key stakeholders include Willamette University's librarians, students, and faculty. Librarians need clear insights into which resources are worth investing in as both the university and its students evolve over time. Students rely on the library for the reading and research materials that are essential to their coursework, so it is critical that relevant resources remain available and accessible. Faculty, in addition to having access to the library's resources, benefit

when these e-resources are tailored to student needs and usage patterns. This allows faculty to confidently direct students to the library for materials that support their success in their courses.

If we are able to identify the usage patterns in electronic resources at Willamette's libraries, we will be able to support the librarians in making data-driven decisions to optimize and ensure that resources across different academic departments are tailored, relevant, and accessible to students across disciplines.

Background & Related Work

Like many other institutions, libraries have navigated changes in usage and funding landscapes over the years. According to the Library Journal article, "Shifting Sands - Budget and Funding 2026" by April Witteveen, many libraries have moved towards supporting more e-content and digital materials as a result of growing interest and usage of digital materials[1]. At the same time, Witteveen notes that while increases in digital materials spending may appear as if it would lead to greater availability of digital materials, spending increases are likely a result of the rising costs of digital licensing from publishers.

Licensing costs of digital materials have led many public libraries to adjust their usage policies to manage rising costs of digital collections. For example, Multnomah County public libraries in Oregon have decreased the number of digital materials patrons are allowed to borrow despite growing interest in digital materials since 2025 [2]. According to Multnomah County libraries, this is due to several factors, including:

- **Consumer vs Library Cost** - Each digital item costs more for a library to purchase than for the consumer. For example, Multnomah County writes that an e-book copy of the 2024 Kristin Hannah title, *The Women*, has a consumer cost of \$15, while a single library copy costs \$60 (see **Image 1**).
- **Purchasing of Multiple Copies** - In addition to the high costs associated with library copies of digital materials, libraries often purchase multiple copies of each digital material to meet the demand of patrons, adding to the costs of materials.
- **Licensing Terms** - Publishers allow libraries to lease materials based on time or total number of checkouts. Libraries must also renew each material to keep them in their library.

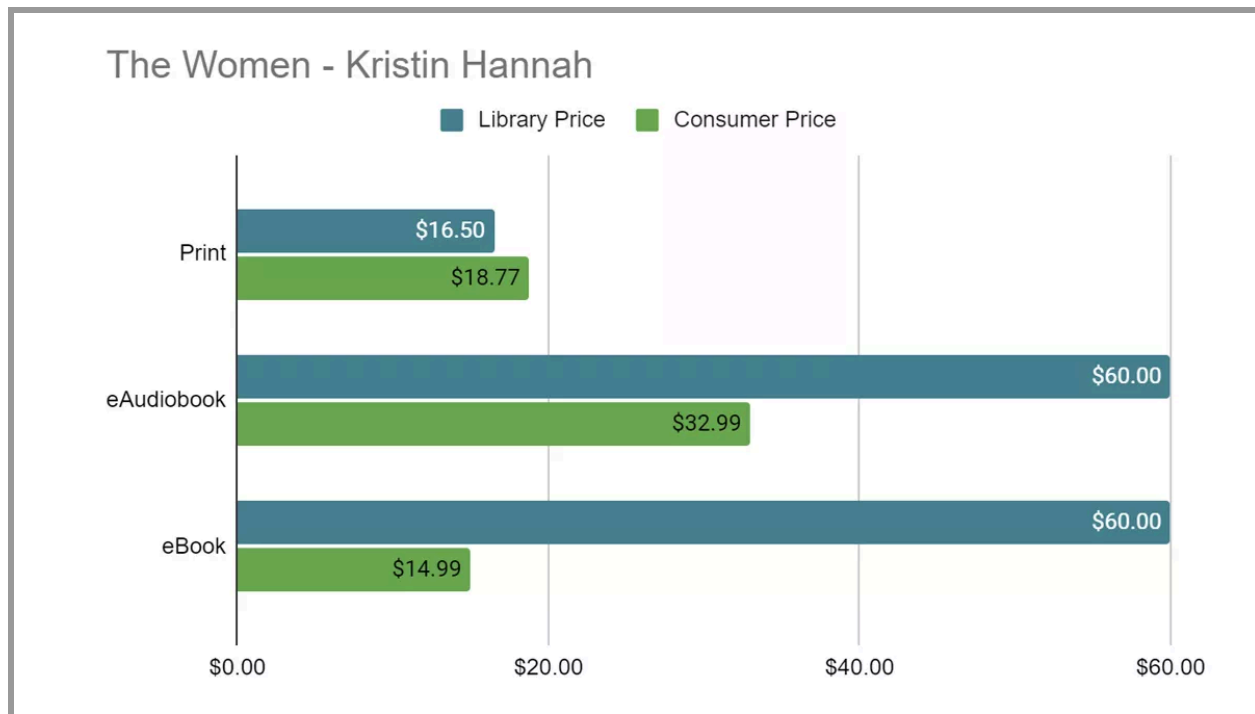


Image 1. Comparison between Library and Consumer costs of the book, The Women[2]. This bar graph shows the costs for Print, EAudiobook, and eBook versions of Kristin Hannah’s The Women (2024), separated between Library Price and Consumer Price. The library price is often higher than the consumer price, with digital materials such as EAudiobooks and eBook versions showing the largest disparity in costs between the consumer and libraries.

Rising costs of maintaining digital collections have led libraries like Multnomah County to reevaluate usage policies to maintain and optimize the use of their digital collections. While Willamette’s libraries have an academic focus, with digital materials and research databases geared toward university students, they face similar challenges to optimize the cost and usage of digital collections. To understand and navigate these questions, data can provide insight into student academic demographics and usage of different materials over time, helping identify inefficiencies in maintaining Willamette’s digital materials.

Data as a Tool to Understand Library Usage

Libraries and library associations across the United States have long advocated using data to understand library usage. In fact, the American Library Association (ALA) has established the Data-Driven Librarianship toolkit to capture key tools for libraries to support planning, improvement, and advocacy. Tools include the ‘Impact Survey,’ which allows libraries to survey patrons’ technology needs, as well as the Public Library Data and Performance Measures to identify key metrics and performance resources to assist with decision-making [3]. Furthermore, the Institute of Museum and Library Services (IMLS) has developed a dashboard using similar metrics developed by the ALA to provide librarians across the country a way to explore and

compare circulation of materials and funding [4]. Regarding digital materials and research databases, the South Central Library System in Wisconsin utilized similar e-resource datasets to create a dashboard that outlines the usage of the library's online resources over time [5]. These efforts indicate a broader interest in utilizing data regarding usage of materials to guide decision-making for libraries across the United States on both a federal and local level.

Data as a Tool to Understand Willamette's Libraries

While this project falls in a similar domain as the previously mentioned public library data projects, it stands out as it integrates higher education contexts and enrollment data to understand a university's library usage. This gives us the opportunity to implement features from existing library data projects to understand the needs of a university's libraries. Furthermore, Willamette currently does not have an existing data pipeline to navigate and analyze library e-resource usage. This uniquely positions our project to develop a specified data pipeline for Willamette's libraries to make sense of usage data to increase efficiencies in the maintenance of the library's resources. The inclusion of these facets provides an opportunity for the project to streamline the available e-resources for students and have a clear impact on student education at Willamette University.

Our output will give librarians data-driven insights into which resources are used most and why. It will provide a better idea of which resources to keep or retire, while ensuring students are aware of the materials available to them. Similarly, this provides an opportunity to make progress on the development of a data pipeline to store, analyze, and develop insights on library e-resource usage data that can be replicated for future school years. The findings and methods utilized in this project will support data-driven decisions for Willamette's libraries in the short term, while also supporting the development of a framework for future analysis of higher education library data.

Data & Data Engineering

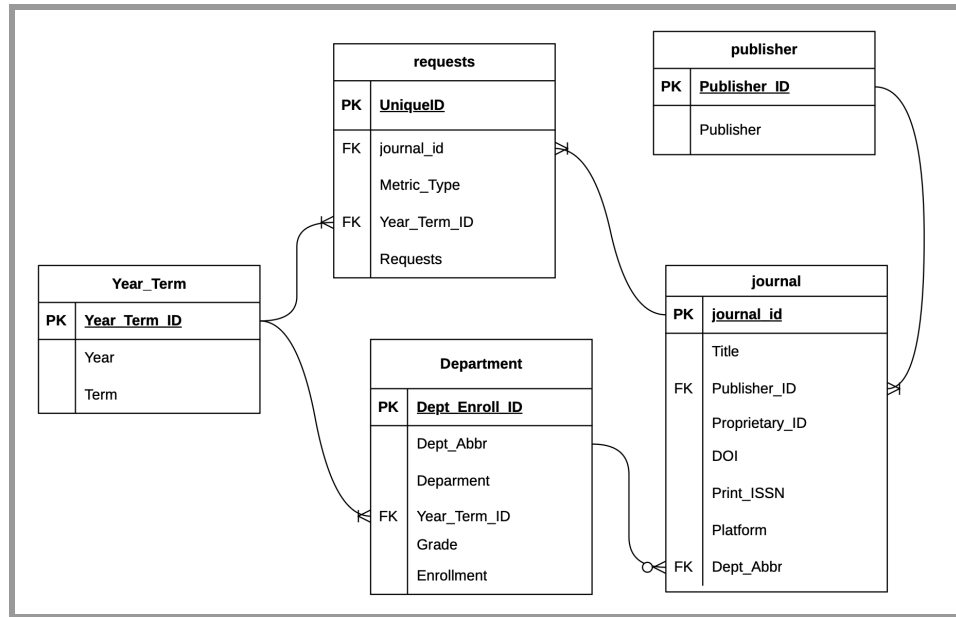
This project combines data from 2 different sources: Willamette Library Usage Request Reports and the Willamette University Factbooks. While the data was fairly clean and complete, there was some work to do for both data sources before it was possible to download it. We needed to label each journal with its subject and write a script to convert the enrollment information into a digestible file type. A quick note: we use the terms "major" and "department" interchangeably here. While some majors are their own independent department, others are not. For instance, there is no standalone STAT department. Statistics instead falls under the School of Computing and Information Sciences (SCIS) umbrella of majors. Because we are interested in analyzing the area of study rather than the technical department each major belongs to, we distinguish between majors individually rather than by the department. Throughout this write-up, "major" and "department" should be understood as referring to the same underlying grouping.

A contact at the Willamette Library supplied the Willamette Library usage requests data, which included 7 spreadsheets from 5 publishers: Oxford University, Sage, Springer Nature, University of Chicago, and Taylor & Francis. Each of these contained monthly data for the number of unique journal requests and the total number of journal requests. The unique item request only counts how many unique journals or articles within each journal were requested. Ultimately, we used the unique requests metric for our analysis. This data spans from July 2023 to March 2026. To analyze the relationship between enrollment in specific majors and journal requests in the same subject area, we needed to classify each journal by subject. This was a large endeavor since there were around 2,000 journals listed. Additionally, because there is some nuance as to what subject each journal belongs to, journals were labeled with their closest related subject based on our discretion, which was then checked for accuracy by a Willamette librarian. The data was then all compiled into one file, ensuring that all of the month columns lined up, as different publishers have data for slightly different time periods. Since we are curious about looking at enrollment and the enrollment data is only as granular as each term, we summed each month for the journal data to also only consider the term. To do this, we needed to assign each month to a term, which we did as follows:

Term	Months
Fall	August-December
Spring	January-April
Summer	May-July

The enrollment data is publicly available on Willamette University's website and contains each semester's enrollment for every major broken down by sex and year in school. We chose this time span to match the time for which we have journal requests data. This was from Summer 2023 to Fall 2025 (Spring 2026 data was not yet available at the time of our data ingestion). The data was stored as an .html on the website, and we downloaded it as a .pdf. Before ingesting this data, we needed to write a script to read the data from the .pdfs to .csvs. We then combined all of this data into one file for ease in ingestion. After this was done, we were able to ingest the data and then wrangle it. The important wrangling steps included summing the male and female data together because we did not differentiate by sex in our analysis. We do not have enrollment data for the summer semesters, so we used each summer's enrollment data from the previous spring, assuming that very few (if any) students declare their major in May through July. Once these steps were completed, the data was ready to be transformed into the new schema.

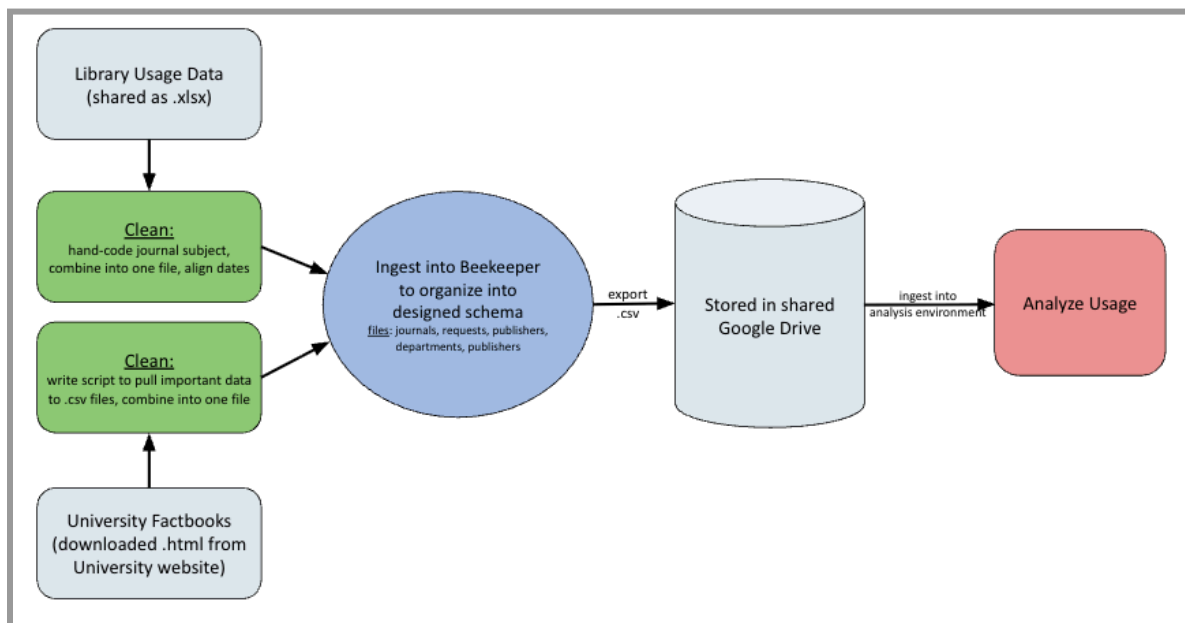
This data was downloaded and ingested into a PostgreSQL database, where it was placed into staging tables. The new tables were then created, and the data was inserted from the staging tables into the new organized tables, which contained the appropriate foreign and primary keys to ensure ease in using joins for our analysis. Our organized schema can be seen below:



Once the data was in this format in our Postgres database, we exported each of the tables into .csv files so we could download them into our analysis environments. These were stored in a shared Google Drive, which allowed us to easily perform our analysis asynchronously without having to re-wrangle the data every time we returned to our coding environment.

Pipeline

The data pipeline described above is visualized below:



Data Ethics

Our data raises few privacy concerns because it does not represent individual people. The journal usage reports are COUNTER TR_J1 reports, which record counts of article requests at the journal level. They contain no user information, no record of who accessed a resource, and nothing that could reidentify a student, faculty, or staff member. Because these files contain no personally identifiable information, there is nothing to redact and no consent process required.

The enrollment data comes from Willamette's public Fact Books, which report aggregate counts of declared majors broken down by grade and sex. These are institution-level summaries, not student records, so they also contain no personally identifiable information. Both datasets are cleared for public use. No files in this project are access-restricted, and both the journal and enrollment data can live in our public repository.

Responsible Use:

The main responsible usage concern is fairness rather than privacy. Our analysis can influence which resources a department keeps or loses, so we are careful not to disadvantage smaller or underserved departments. We compare journals within each department rather than only across the whole collection, and we strive to make equitable suggestions backed by our findings. Because we do not want to cause harm to any department, we treat our prioritization output as suggestions for librarian review, not automatic decisions. Final decisions and actions rest with the Willamette librarians, who have strong domain knowledge in this subject matter and can best determine what resource allocation will best fit the needs of Willamette students and the broader Willamette community.

Methods & Evaluations

Approach

We use exploratory analysis and descriptive statistics rather than predictive modeling. The stakeholder question is descriptive, identifying which resources are being used, by whom, and how usage is changing over time. A predictive model answers a question the librarians did not ask, and with under three years of monthly data, there is not enough history to train or validate one.

Usage is measured with COUNTER Unique Item Requests rather than Total Item Requests. Total Item Requests counts repeated views of the same article, and different publisher platforms inflate that count differently depending on how their interface serves PDFs and HTML full text. Unique Item Requests count each article once per session, which allows for meaningful comparisons across publishers.

Journals are grouped into five publisher packages: Springer, SAGE, Taylor & Francis, Oxford, and University of Chicago. The library subscribes at the package level rather than by individual title, so the package is the unit at which a keep or cancel decision is made. The raw files contain nine distinct platform values because certain publishers report across multiple platforms. SpringerLink, Springer Nature Link, and nature.com are all Springer content and were collapsed into a single Springer package. A small number of records listed ResearchGate as the platform, but these are not part of any library subscription and were dropped.

Requests are aggregated to the term level to match the enrollment data, which is only published per term. For department-level comparison, we normalize by enrollment, reporting unique requests per enrolled student rather than raw request counts, so a large department does not automatically appear more engaged than a small one.

We could not answer one part of RQ1. The COUNTER reports contain no course or class identifier, and neither does the enrollment data, so we cannot link a journal request to a specific class. We report on subject, department, and term, without attempting to analyze individual courses.

Validation

There is no ground truth for what “correct” usage looks like, so validation focused on two things: whether the data supports the counts we report, and whether a finding survives reasonable alternative choices.

Data integrity. Oxford and Springer each supplied two export files covering overlapping time periods rather than one continuous file. Concatenating all seven spreadsheets double-counts every journal and month that appears in both files, which inflates total requests by roughly 4.4%. We duplicate at the journal and term level, taking the maximum value where the two files disagree, before aggregation.

Reporting floor. Publisher COUNTER reports omit titles that received no requests in a given period. A blank cell therefore means “not reported,” rather than “zero requests,” and the minimum observed value in the data is one request, rather than zero. This collection cannot be described in terms of unused journals, and any month-over-month comparison across the full title list would confuse a change in reporting with a change in usage. For the time series, we restrict to the 1,493 journals that appear in every month of the window, so the panel is constant and the trend reflects usage instead of coverage.

Coverage gaps. March 2026 appears in the data for University of Chicago only. Including it would show an apparent collapse in usage that is rather an artifact of four publishers not having reported yet. That month is excluded from all time series, setting the analysis window from July 2023 through February 2026.

Robustness. The year-over-year decline was checked separately within the publishers that supplied a single continuous file. Thus, we confirm it is not an artifact of the deduplication step, and it appears in each of them rather than in one alone. Where a result depends on a chosen cutoff, we tested it under several cutoffs and report it only where the conclusion holds across them. An earlier version of the prioritization view used a consistency axis alongside volume, which we dropped after finding the two were highly rank-correlated (Spearman's rho of about 0.95) and were therefore measuring the same thing twice. The current version uses recency share rather than an arbitrary midpoint.

Results & Visualization

Figure 1: Journal Usage by Month, 2023-2026

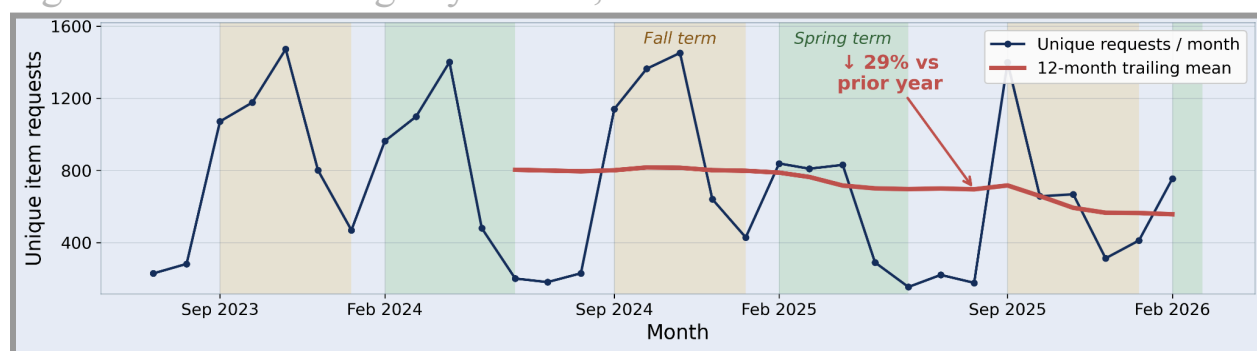


Figure 1. Journal usage by month, July 2023 to February 2026. Unique item requests per month, summed across the 1,493 journals reporting in every month of the window ($n = 1,493$). The red line is the 12-month trailing mean. March 2026 is excluded because only one of five publishers reported that month.

Figure 1 shows total unique item requests per month from July 2023 through February 2026, restricted to the 1,493 journals that reported in every month of the window ($n = 1,493$ journals). The vertical axis represents unique item requests per month across the whole collection. Shaded bands mark Fall and Spring terms.

Usage tracks the academic calendar closely. Requests peak in the middle of each term, fall off sharply during winter and summer breaks, and reach their yearly low in June and July. Fall consistently produces the highest sustained usage of the three terms.

There appears to be a clear direction in this trend. The twelve-month trailing mean fell from roughly 787 unique requests per month to roughly 556, a decline of 29% over the most recent year. Because the panel of journals is held constant, this is not a reporting gap. The decline also appears within the publishers that supplied a single continuous export file, which rules out the deduplication step as a cause. We report the decline as real and note that this analysis cannot identify its cause. The rise of large language models as a research tool is one plausible explanation, but this is speculation and not directly supported by this data.

The isolated spike in September 2025 is driven by a single newly activated journal title rather than a broad return to earlier usage levels, and the series returns to trend the following month. This should not be interpreted as a recovery.

Figure 2: Departments Ranked by Usage Intensity per Student

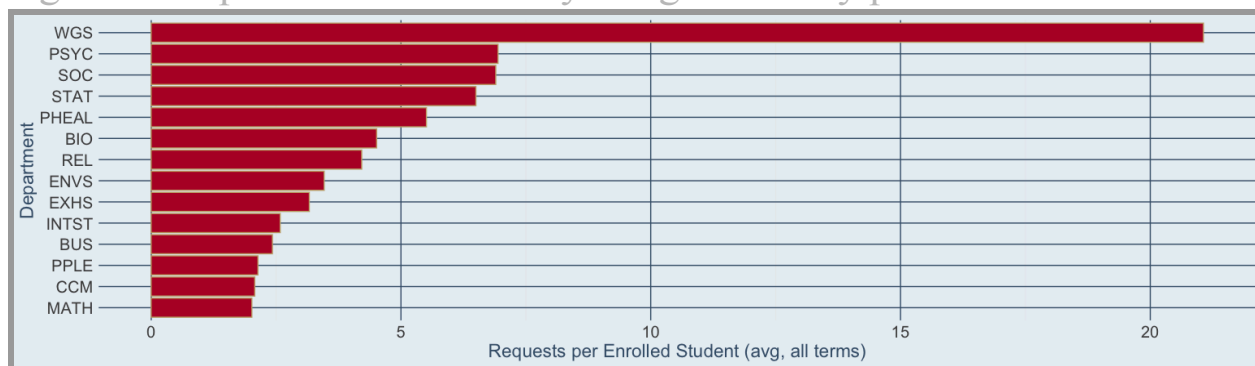


Figure 2. Journal requests per student broken down by department. Total unique item requests for each department divided by the average enrollment over all terms of available enrollment data.

Figure 2 shows the number of journal requests per student within each department, averaged across all terms present (Summer 2023-Fall 2025). The vertical axis displays the departments with the top 14 requests per enrolled student. The horizontal axis denotes the number of requests per student.

Women’s and Gender Studies (WGS) is a prominent outlier, with an average of 3.88 majors per term, but 957 total requests. While 957 is not necessarily a large number of requests (9th overall), the low enrollment makes this notable. While there is no obvious reason for why this might be, we hypothesize that it could be a mix of (1) popular topics for research in WGS journals, which mostly focus on sexuality, gender identity, or women’s studies; (2) classes in other departments, like PNCA’s liberal arts, using WGS-labeled journals; or (3) use by other campus organizations like the Gender Resource & Advocacy Center (GRAC).

Psychology (PSYC) and Sociology (SOC) follow WGS with the next highest requests per enrolled student ratio, with averages of 96.12 and 36.75 students per semester, and 5339 and 2028 total requests, respectively. These departments have higher enrollment and a larger volume of journals, and both rank in the top 5 for journal count. Public Health (PHEAL), Biology (BIO), and Environmental Science (ENVS), which make up the rest of the top 5 departments by journal count and also have consistently high enrollment, are also notable.

Statistics (STAT) ranks fourth in requests per student, which supports the notion that statistical methods are used across many other disciplines. The same applies to all other departments, supporting our hypothesis that requests came from a mix of students from different departments and other members of the Willamette community. Overall, this graphic shows that

enrollment is not a strong indicator of journal usage, given the variation and the volume of out-of-major journal usage by students, faculty, and staff alike.

Figure 3: Usage Requests by Enrollment

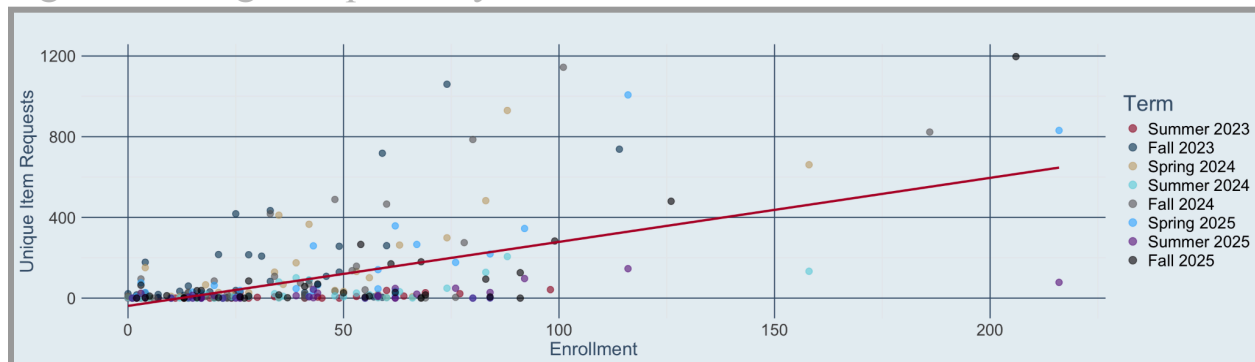


Figure 3. Journal requests per term broken down by department compared against that major's enrollment. Total requests aggregated each department and each term. Line of best fit trends upward with a slope of 3.2 and an intercept of -38.

Figure 3 shows the relationship between enrollment and usage requests, total for each term and broken down by department/major. While there is a high amount of noise in this graphic, the upward trend remains noteworthy.

The slope of the line of best fit is 3.173, meaning that for each additional student enrolled in that major, we expect journal usage to increase by an average of 3.173 requests. The slope was statistically significant ($p < 0.001$), indicating strong evidence that the relationship between enrollment and journal usage requests is not likely due to chance. The intercept (-38.61) is unexpected in the context of our problem. It suggests that an enrollment of 0 in a given term would be expected to have a negative number of usage requests, which is not possible.

The Pearson's correlation coefficient is 0.572, which indicates a strong positive correlation between enrollment in a particular major for a given semester and requests for journals in that major in the same semester. Although we cannot conclude from this that higher enrollment causes higher usage, this relationship is present and notable.

The R-squared statistic is 0.3275, meaning that about 33% of the variation in the usage requests can be explained by enrollment. The remaining ~67% of variation is likely due to other factors that were discussed alongside **Figure 2**, such as out-of-major requests, having journal-heavy classes, or usage by others in the Willamette community.

While this overall trend is strong, there is substantial variability around the line for individual department-term combinations. The residual standard error is 159.7, meaning actual usage requests for a given department in a given term can differ from the model's prediction by a wide margin, which is consistent with the visible scatter in Figure 3.

Figure 4: Usage Requests by Publisher/Package

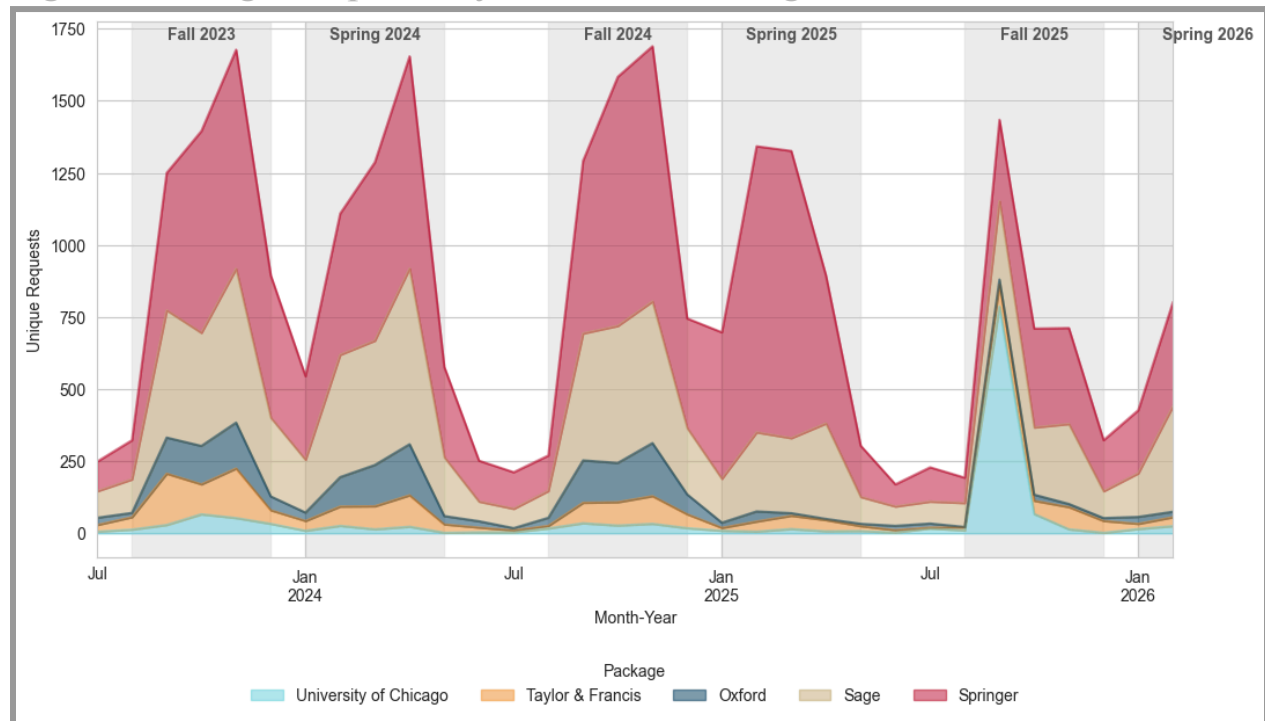


Figure 4. Share of requests by publisher/package by month, July 2023 to February 2026. Unique item requests per month summed across all 2,126 journals (includes all journals that did not report in every month of the window, $n = 2,126$). March 2026 is excluded because only one of five publishers reported that month.

Figure 4 demonstrates the share of unique journal requests by month. The horizontal axis indicates the month and year, while the vertical axis indicates the number of unique requests. The filled colored spaces represent the share of journal requests by each package/publisher. Background enclosures indicate the Fall and Spring semesters.

Similar to **Figure 1**, usage mirrors the academic calendar, with requests peaking in the middle of each term and falling sharply during academic breaks. When disaggregated by publishers, Springer and Sage consistently have the largest share of usage requests. Springer and Sage are followed by Oxford and Taylor & Francis, with University of Chicago consistently making up the smallest share of requests for each month.

September 2025

While University of Chicago typically represents the smallest share of journal requests each month, its share notably spikes in Fall 2025. Upon closer inspection, there is an unusual spike in requests for the journal *Environmental History* (774 unique item requests as opposed to usual requests ranging from 0-20). *Environmental History* is a journal covering research that explores the relationship between humans and the environment. Research included in this journal typically intersects geography, anthropology, natural sciences, science and technology studies,

political ecology, and other disciplines[6]. According to Willamette’s course catalog, several courses are typically offered in the Fall that align with this topic area or may impact journal request usage[7]. Courses include:

Course #	Title	Summarized Course Description
ENVS 340	Biogeochemistry	Explores how human behavior and Earth’s history impact biogeochemical cycling using primary literature. <i>Offered Fall of Odd-Numbered Years.</i>
ENVS 328	Health & the Global Environment	Examines how geography has been used to understand disease patterns and outbreaks, access to health care, health inequality, and the impact of the environment on health. <i>Offered Fall, every year.</i>
ENVS 323	Environmental Law, Regulation	Examines the pervasive and fundamental role the law, courts, and regulatory agencies play in the interpretation, implementation, and enforcement of environmental regulation. <i>Offered Fall of Odd-Numbered Years</i>
ENVS 304W	Politics of Environmental Ethics	Analyzes human/nature relationship and its impact upon political theory and ethics. <i>Offered Fall every year.</i>
ENVS 120	Social Systems, Environment	Examines the effects of human actions and social systems on the natural world. Emphasizes science and social-science based approaches to understanding environmental problems. <i>Offered Fall and Spring every year.</i>
ENVS 495W	Senior Capstone	Provides students the opportunity to cultivate professional work habits necessary for success by applying and integrating skills and knowledge via a semester-long investigation of a topic. Students will focus on writing a literature review and conceptualizing research. <i>Offered Fall every year.</i>

Table 1. Table of Environmental Science Courses offered in the Fall, in which students may be researching topics that are covered by the journal, *Environmental History*.

Crosswalk of Publisher, Unique Journals and Departments

As mentioned previously, Springer and Sage typically have the largest share of unique journal requests each month. This is likely because Springer and Sage have the largest count of unique journals that have ever been requested from their platforms. However, in terms of representation of academic departments, Taylor & Francis appears to have the largest number of unique departments represented within this dataset, at 34 departments of the 55 departments that have been included in our analysis. Sage and Springer follow suit at 31 and 26 departments, respectively (see **Table 2**).

Publisher	# Unique Journals Requested	# Unique Departments	Top 3 Departments Represented in Publisher by Number of Journals <i>Department (Number of Journals)</i>	Top 3 Departments by Number of Requests <i>Department (Number of Requests)</i>
Springer	926	26	1. Biology (n = 198) 2. Public Health (n = 127) 3. Environmental Science (n = 111)	1. Environmental Science (n = 3552) 2. Psychology (n = 2625) 3. Biology (n = 1813)
Sage	616	31	1. Psychology (n = 93) 2. Sociology (n = 87) 3. Politics, Policy, Law, and Ethics (n = 68)	1. Psychology (n= 2530) 2. Sociology (n = 1191) 3. Politics, Policy, Law, and Ethics (n = 774)
Taylor & Francis	336	34	1. Environmental Science (n = 31) 2. Politics, Policy, Law, and Ethics (n = 31) 3. Biology (n = 29)	1. Math (n= 327) 2. Women's and Gender Studies (n = 125) 3. Environmental Science (n = 124)
Oxford	205	23	1. Biology (n = 72) 2. Public Health (n = 28) 3. Politics, Policy, Law, and Ethics (n = 19)	1. Biology (n = 679) 2. Public Health (n = 284) 3. History (n = 182)
University of Chicago	43	14	1. Sociology (n = 6) 2. Economics (n = 6) 3. Environmental Science (n = 5)	1. Environmental Science (n = 849)* 2. Sociology (n = 103) 3. Biology (n = 85)

Table 2. Crosswalk of publisher journal count and representation of departments by each publisher. The columns “# Unique Journals” and “# Unique Departments” are colored on a gradient with the darkest shade indicating the highest value for that column and the lightest shade indicating the lowest value for that column. **NOTE:** requests for Environmental Science journals from University of Chicago include 774 requests from September 2025. If September 2025 is omitted, n = 75. The ranking would then be: Sociology (n = 103), Biology (n = 85), and Women and Gender Studies (n = 83)

Additionally, the top 3 departments represented within each publisher in this dataset have been indicated. Across publishers, there are common departments for the top 3 represented departments based on journal count. For example, both Environmental Science (ENVS) and Politics, Policy, Law, and Ethics (PPLE) appear in the top 3 for at least 3 publishers. This aligns with usage because these departments appear in the top 10 for usage intensity per student enrolled in the major (see **Figure 2**). Furthermore, for Springer, Sage, Oxford, and University of Chicago the top 3 represented departments generally align with the top 3 departments based on requests, meaning that availability/representation and usage align. Taylor & Francis shows the most discrepancy between the top 3 represented departments by journal count and the top 3 based on journal requests. While Taylor & Francis’ top 3 represented departments based on journal count are similar to the other publishers, the top 3 most requested departments include Math (MATH), Women’s and Gender Studies (WGS), and Environmental Science (ENVS).

Similarly for the University of Chicago, when the unusual spike for Environmental Science materials in September 2025 ($n = 774$) is removed, the top 3 requested Departments are Sociology ($n = 103$), Biology ($n = 85$) and Women and Gender Studies ($n = 83$). On further analysis, this discrepancy between department representation based on journal count and student requests can also be seen across the whole dataset.

Figure 5: Top 20 Departments by Count of Unique Journals and Total Unique Requests

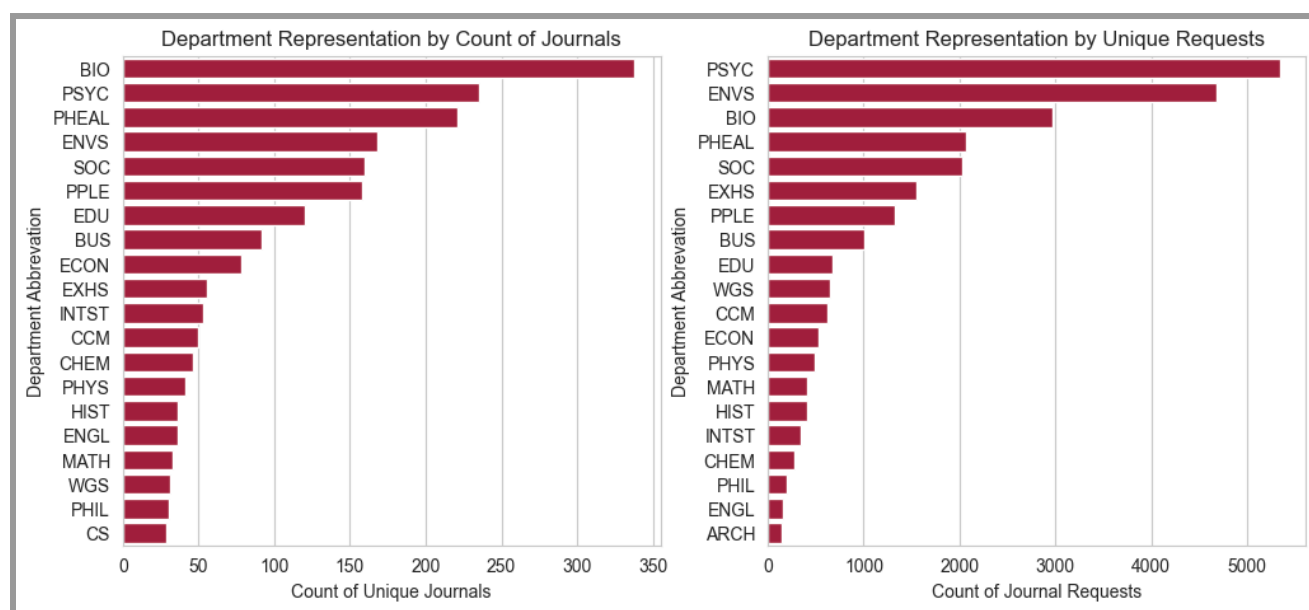


Figure 5. Top 20 departments represented in the dataset based on count of unique journals or total unique journal requests.

Figure 5 ranks the top 20 departments represented within this dataset based on the count of journals and total unique journal requests. The left most figure ranks departments by the count of unique journals categorized under that department. The right most figure ranks departments by total unique requests. The horizontal axis represents the metric for ranking (Count of Unique Journals or Unique Journal Requests). The vertical axis indicates the department.

The top 5 departments remain consistent in terms of representation ranking across both metrics. However, several departments show some discrepancy between count of journals and usage. These include:

- **Women's Gender Studies (WGS):** By count of journals, WGS was ranked 18th, while in terms of requests, WGS is ranked 10th.
- **Computer Science (CS):** By count of journals, CS is ranked 20th. By requests, CS does not make the top 20.
- **Chemistry (CHEM):** By count of journals, CHEM is ranked 13th. By requests, CHEM is ranked 17th.
- **Exercise Health Science (EXHS):** By count of journals, EXHS ranks 10th. By requests,

EXHS is ranked 6th.

- **Math (MATH):** By count of journals, MATH ranks 17th. By requests, MATH ranks 14th.

Discussion, Recommendations & Limitations

Discussion

RQ1: What are the patterns in usage with regard to subject matter, departments, semesters, and classes?

Analysis of overall usage patterns across terms indicates that, while usage of e-resources often mirrors the semester term, it appears to be declining overall. While this report does not identify the cause, speaking with our stakeholders has suggested that shifting student research patterns, such as the rise of large language models and the de-prioritization of journal materials as a resource for research, may contribute to this overall decline. While there are no definitive reasons that can be derived from our existing data and analysis, these usage patterns and discussions represent a larger need to understand shifting student research habits and classroom practices regarding what resources are discussed and promoted.

Regarding subject matter and academic department, we find a strong positive correlation between the enrollment in a particular major in a given semester and the requests for journals in that major in a given semester. Departments, such as Women's and Gender Studies (WGS), were a significant outlier when considering usage intensity (i.e., requests per student). Our existing hypotheses for the underlying reasons behind this outlier include 1.) the popularity of the topic as a whole, 2.) the cross-disciplinary use of these journals across majors and schools, and finally, 3.) usage from other campus organizations, such as the Gender Resource & Advocacy Center (GRAC). These findings indicate that a higher enrollment number for a particular major in a given semester may indicate a higher usage of journals categorized under that major.

In terms of semester courses, we see that there is a potential relationship between course offerings and journal requests. In particular, when an unusual spike in Fall 2025 for the journal *Environmental History* (published by University of Chicago) is crosswalked with the university course catalog, it is possible to identify several classes that overlap in terms of topic area, such as a senior capstone in which students are focused on developing literature reviews and research.

When considering department representation across publishers, publishers with more unique journals, such as Springer and Sage, tend to make up a larger share of journal requests. However, publishers like Taylor & Francis provide more variety across departments. Willamette's availability of E-Resources for departments such as Psychology, Biology, Public Health, and Environmental Science appears to match the need in terms of requests. Some departments, such as WGS and MATH, appear to rank higher in requests than in terms of journal availability. Within this dataset, WGS and MATH requests tend to be fulfilled by Taylor &

Francis, even if Taylor & Francis appears to have more journals aligning with PPLE or BIO topics.

RQ2: Can we use the patterns in usage to make suggestions on which resources to prioritize, and which resources are not as important?

The analysis of usage patterns based on semesters, departments/subject matters, course offerings, and publisher revealed several strong patterns. These patterns indicate changing resource usage, so our suggestions link student research behavior, enrollment, and publisher offerings to support navigating these shifts.

Our Recommendations

Recommendation 1: Behavioral Perspective- Promote Library E-Resources Early in the Semester

Findings from this analysis indicate an overall decline in usage of e-resource materials. All current hypotheses connect back to student research habits. Considering that usage often peaks at the midpoint of each semester, Willamette's teaching librarians can take advantage of the early semester weeks to visit classes that heavily rely on academic research. This includes capstone courses, capstone prep-classes, and other research/literature review-heavy courses. These visits allow librarians to promote the e-resources and demonstrate how e-resources at Willamette's libraries can be utilized prior to usage peaks in the semester.

Recommendation 2: Enrollment/Discipline Perspective - Consider Enrollment per semester while taking note of overall campus topic interests

As enrollment for a major during a given semester increases, usage requests for journals relating to that subject also increase. This is a feature to consider when identifying which topic areas/journals students may request most and which are important to invest in. Outliers, such as the Women's and Gender Studies major, where enrollment is low but requests per student are high, present another feature to consider. We hypothesize this discrepancy is due to the popularity of WGS at Willamette and its interdisciplinary use across multiple majors and schools. Willamette's librarians should consider enrollment within academic disciplines but also consider which topics are known to be interdisciplinary and generally popular across campus.

Recommendation 3: Publisher and Journal Perspective - Consider resources for Women's and Gender Studies (WGS)

When considering publishers and representation of journals across departments, Springer and Sage tend to have a larger collection of unique journals, but Taylor & Francis, though smaller by total count of journals, has a higher number of departments represented. Furthermore, analysis shows that WGS is unique as it is ranked lower in representation of journals, yet ranks higher in terms of requests, and requests per student enrolled in the department. Furthermore, for Taylor & Francis, WGS (along with Math) appears to be one of the most requested topics/departments. This may indicate that, at the least, resources for WGS should be considered.

Taylor & Francis should also be considered because it represents a wider variety of departments and seems to fulfill requests for WGS materials.

Limitations

This project analyzes the requests per student. However, the current e-resource usage data includes journal access from everyone in the Willamette community, not just students. This potentially inflates usage for each academic discipline. Similarly, our existing journal data does not reflect/delineate between journal requests submitted by out-of-major students and major students, potentially impacting our findings. Finally, many journals don't fit neatly into one subject, making it difficult to truly assess how usage differs across subjects/academic disciplines.

Future Work

In the Results & Visualizations, we provide several hypotheses for the patterns in usage. We emphasize that this analysis does not confirm causation. Thus, these hypotheses require further research to fully understand and confirm their validity. Further research will allow Willamette's librarians to better understand patterns that this analysis reveals and adjust practices and tailor resources to the needs of Willamette's students. Potential research questions include:

1. What resources are students using to conduct research?
2. Are these resources available at the Library or are students relying on outside resources?
3. What tools are students exposed to for research in the classroom?
4. What topics are students typically interested in?

Given our existing analysis and limitations of the data, we recommend utilizing course enrollment data to explore how enrollment in semester courses and the structure of semester courses (research-heavy courses vs. non-research-heavy courses) may impact e-resource usage. However, we do recognize that enrollment data may contain Family Educational Rights and Privacy Act (FERPA) concerns and, as a result, may not be as easily accessible. More broadly, because this project is the first iteration of a data process to ingest, clean, and visualize data from several sources of higher education data, we recommend continuing to refine this process. Refinement could include developing clearer/more concise categories of journals and building an accessible/easily replicable process for non-technical audiences to develop visualizations/analysis on a regular basis.

Conclusion

This project has presented an opportunity to examine how contextual factors such as student enrollment and subject matter can impact the popularity and usage of e-resources. This presents a consequential research project given its timeliness during a time of changing library

usage patterns and the necessity to optimize library resources to the needs of patrons. Our analysis indicates that overall e-resource usage is trending downwards, while analysis of enrollment shows a positive relationship between a given major and journal requests for that topic area. Throughout this trend, there are also outliers such as Women's and Gender Studies, in which there is a low enrollment but high usage. These findings lead to our recommendations to utilize early semester weeks to do outreach in order to promote library e-resources, review enrollment data as a potential feature for understanding usage patterns in a given semester, and finally, consider resources for outlier topics such as Women's & Gender Studies that have a high usage despite a low enrollment. We also present suggestions for further research to identify the underlying reasons for the patterns revealed in this analysis. This project reveals patterns in a changing library/research landscape. It also highlights the strengths and opportunities of data as a tool to understand these patterns and identify recommendations to tailor resources while maintaining fairness between departments across Willamette's campuses. This presents data as a tool to support Willamette Library's goals of being good stewards of research to current and future students.

Reproducibility

Repository. All code, the normalized data files, and the notebooks that produced the figures in this report are at <https://github.com/sophiarabb/data510-libraries>. Both the journal usage data and the enrollment data are cleared for public posting, so the repository is complete and requires no restricted files to run.

Data files. The analysis reads five normalized CSVs exported from our PostgreSQL database: publishers.csv, requests.csv, journals.csv, departments.csv, and year_term.csv. These correspond to the schema shown in the Data and Data Engineering section.

Shared loader. A shared module, library_data.py, handles loading and the two transformations that must happen before any aggregation: deduplication of the overlapping Oxford and Springer export files, and collapsing the nine raw platform values into five publisher packages. All three team members load the data through this module so that the corrections are applied identically across our analyses.

Environment and dependencies. Python 3.13, pandas for analysis, pdfplumber for extracting enrollment tables from the Fact Book PDFs, and matplotlib for figures. PostgreSQL 16 for the database, accessed through Beekeeper Studio. No machine learning libraries are used, since the analysis is descriptive.

To reproduce. Clone the repository, install the dependencies listed in requirements.txt, and run the notebooks in the analysis folder in order. The notebooks read from the CSVs in the data folder and regenerate every figure in this report.

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